

Project Documentation

CS2704

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Introduction and Background

This project investigates whether there is a direct relationship between the number of EV charging stations available in each U.S. state and the growth in EV registrations from 2022 to 2023. The data being used is accessible publicly, this study aims to assess if investments in EV infrastructure have a significant impact on adoption rates or other factors are more influential.

Hypothesis

Original: States with more EV charging stations would experience higher growth in EV adoption.

Final hypothesis is the same as the original. We tested it out by analyzing the state-level data on the number of charging stations and changes in EV registrations between 2022 and 2023.

The analysis and the implication

Three datasets were merged for better analysis. A calculated EV growth percentage was used as the response variable. Descriptive analytics showed that the average EV growth across states was approximately 47%. Visualizations used (histogram, scatter plot, heatmap) revealed that although charging station count is strongly correlated with the total numbers of EVs, it has almost no correlation with EV growth percentage.

Linear Regression analysis confirmed the lack of significant relationship, the coefficient was extremely weak, and p value was very high, and R^2 value was almost zero.

The implication is that EV adoption rate is influenced heavily by other factors, such as government incentives, consumer attitudes, electrical prices.

Conclusion

In this analysis, I explored whether the number of EV charging stations in each U.S. state could predict the percentage growth in electric vehicle registrations from 2022 to 2023. The hypothesis was tested using linear predictive model and the results showed a very weak and statistically insignificant relationship. The findings implicate that while infrastructure is important, it does not directly drive short-term EV growth, and that other factors such as policy incentives, consumer behaviour and prior adoption levels likely to have a much bigger role.

References

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- <https://afdc.energy.gov/vehicle-registration?year=2022>
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